PROBLEM 02

Kinematics, Spiral

Original statement · Problems.pdf, p. 6

Polar Kinematics

A planar spiral is given by the equation $r = \alpha e^{\beta\theta}$, where α and β are constants, r is the radial coordinate, and θ is the angular coordinate. If a particle moves along the spiral according to the equation $\theta = \delta t^n$ (where δ is a constant), determine:

(a) The radial and transverse components of the velocity.

(b) The radial and transverse components of the acceleration.

(c) The tangential and normal components of the velocity.

(d) The tangential and normal components of the acceleration.

(e) The radius of curvature of the spiral.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle P moves along the planar spiral $r = \alpha e^{\beta\theta}$ with the prescribed angular motion $\theta(t) = \delta t^n$.
- **Hypotheses:** $\alpha, \beta, \delta, n > 0$ are constants and $t > 0$. At $t = 0$, $\dot{\theta} \propto t^{n-1}$ diverges for $0 < n < 1$ and $\ddot{\theta} \propto t^{n-2}$ diverges for $1 < n < 2$; for $n = 1$ and for $n \geq 2$ every quantity is finite at $t = 0$ (for $n \geq 2$ the speed vanishes there, which is not a singularity, because $\mathbf{e}_t$ and ρ do not depend on the speed). $t > 0$ is kept for all n for uniformity. The motion is planar ($z = 0$).

- **Constraints and degrees of freedom:** the spiral $r = \alpha e^{\beta\theta}$ is a holonomic, scleronomic constraint in the plane. The particle then has $\mathrm{CDOF} = 2 - 1 = 1$, and the motion law $\theta = \delta t^n$ fixes it completely.
- **Coordinates:** polar coordinates (r, θ) about the origin O of the spiral; the angle θ is the free coordinate.
- **Parameters and units:** $[r] = \mathrm{m}$, so $[\alpha] = \mathrm{m}$; β is dimensionless; $[\theta] = \mathrm{rad}$, so $[\delta] = \mathrm{rad} \cdot \mathrm{s}^{-n}$.

Reference Frames & Vector Bases

- **Inertial basis:** $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Polar basis:** $\mathcal{B}_{\text{polar}} = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$, a rotation of $\mathcal{B}_0$ by θ about $\mathbf{k}_0$:

$$[{}_0R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_\theta] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_\theta][{}_0R_\theta]^T = I$$

The product RR^T has diagonal entries $\cos^2\theta + \sin^2\theta = 1$ (and 1) and off-diagonal entries $\cos\theta\sin\theta - \sin\theta\cos\theta = 0$.

- **Angular velocity of the basis:** $\boldsymbol{\omega} = \dot\theta\,\mathbf{k}_0$. By Poisson's formula (Notes Sec. 2.6.2, Eqs. 2.25-2.26): $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$.
- **Intrinsic basis from the polar basis** (derived in 3.1(c), where the sign of the normal is justified): $\mathbf{e}_t = (\beta\,\mathbf{e}_r + \mathbf{e}_\theta)/\sqrt{1+\beta^2}$, $\mathbf{e}_n = (-\mathbf{e}_r + \beta\,\mathbf{e}_\theta)/\sqrt{1+\beta^2}$ and $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \mathbf{k}_0$. The components of $\mathbf{e}_t$ and $\mathbf{e}_n$ in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are the columns of

$$[{}_{\text{polar}}R_{\text{tn}}] = \frac{1}{\sqrt{1+\beta^2}}\begin{pmatrix} \beta & -1 \\ 1 & \beta \end{pmatrix}, \qquad \det = \frac{\beta^2+1}{1+\beta^2} = 1, \qquad RR^T = I$$

(rows of norm $(\beta^2+1)/(1+\beta^2) = 1$ and orthogonal, $(\beta - \beta)/(1+\beta^2) = 0$). It is completed with $\mathbf{e}_b = \mathbf{k}_0$ as third row and column, $(0, 0, 1)$, so the full 3×3 matrix is a rotation about $\mathbf{k}_0$. This is a planar rotation by the constant angle $\arctan(1/\beta)$: the tangent makes a constant angle with the radial direction (equiangular spiral).

Derivation

3.1 Kinematics

(a) Radial and transverse components of the velocity

Pedagogical Justification: in polar coordinates $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta$ (product rule and Poisson). The coordinates depend on time through $\theta(t)$, so the chain rule is needed.

$$\dot{\theta} = \frac{d}{dt}\left(\delta t^n\right) = n\delta t^{n-1}, \qquad \frac{dr}{d\theta} = \frac{d}{d\theta}\left(\alpha e^{\beta\theta}\right) = \alpha\beta e^{\beta\theta} = \beta r$$

$$\dot{r} = \frac{dr}{d\theta}\,\dot{\theta} = \beta r\, n\delta t^{n-1} = \alpha\beta n\delta t^{n-1} e^{\beta\delta t^n}$$

$$\boxed{v_r = \dot{r} = \beta n\delta t^{n-1}\, r}, \qquad \boxed{v_\theta = r\dot{\theta} = n\delta t^{n-1}\, r}$$

(b) Radial and transverse components of the acceleration

Pedagogical Justification: the polar acceleration is (Notes Eq. 2.30; product rule applied to $\mathbf{v}$ with Poisson's formula)

$$\mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

- **Second derivatives** (product rule, with $\dot{r} = \beta r\dot{\theta}$):

$$\ddot{\theta} = \frac{d}{dt}\left(n\delta t^{n-1}\right) = n(n-1)\delta t^{n-2}, \qquad \ddot{r} = \beta\left(\dot{r}\dot{\theta} + r\ddot{\theta}\right) = \beta\left(\beta r\dot{\theta}^2 + r\ddot{\theta}\right) = \beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}$$

- **Radial component:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = \left(\beta^2 - 1\right) r\dot{\theta}^2 + \beta r\ddot{\theta} = r\left[\left(\beta^2 - 1\right) n^2\delta^2 t^{2n-2} + \beta n(n-1)\delta t^{n-2}\right] = n\delta t^{n-2}\, r\left[\left(\beta^2 - 1\right) n\delta t^{n} + \beta(n-1)\right]$$

(substituting $\dot{\theta}^2 = n^2\delta^2 t^{2n-2}$ and $\ddot{\theta} = n(n-1)\delta t^{n-2}$, then factoring out $n\delta t^{n-2}$, since $t^{2n-2} = t^{n-2}t^{n}$).

- **Transverse component:**

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\ddot{\theta} + 2\beta r\dot{\theta}^2 = r\left[n(n-1)\delta t^{n-2} + 2\beta n^2\delta^2 t^{2n-2}\right] = n\delta t^{n-2}\, r\left[(n-1) + 2\beta n\delta t^{n}\right]$$

(c) Tangential and normal components of the velocity

> **Pedagogical Justification:** the velocity is always tangent to the path, so in the intrinsic basis $\mathbf{v} = v\,\mathbf{e}_t + 0\,\mathbf{e}_n$ and the normal component is identically zero.

$$v_t = v = \sqrt{v_r^2 + v_\theta^2} = r\dot{\theta}\sqrt{\beta^2 + 1} = n\delta t^{n-1}\, r\sqrt{1 + \beta^2}, \qquad v_n = 0$$

- The unit tangent follows from $\mathbf{v} = r\dot{\theta}\left(\beta\,\mathbf{e}_r + \mathbf{e}_\theta\right)$: $\mathbf{e}_t = \left(\beta\,\mathbf{e}_r + \mathbf{e}_\theta\right)/\sqrt{1+\beta^2}$. The normal toward the centre of curvature must satisfy $\mathbf{a}\cdot\mathbf{e}_n > 0$. The left normal $\mathbf{k}_0 \times \mathbf{e}_t = \left(-\mathbf{e}_r + \beta\,\mathbf{e}_\theta\right)/\sqrt{1+\beta^2}$ does: its projection of the acceleration (with a_r and a_θ of part (b)) is $\left(\beta a_\theta - a_r\right)/\sqrt{1+\beta^2}$, and

$$\beta a_\theta - a_r = \beta\left(r\ddot{\theta} + 2\beta r\dot{\theta}^2\right) - \left(\left(\beta^2 - 1\right) r\dot{\theta}^2 + \beta r\ddot{\theta}\right) = \left(1 + \beta^2\right) r\dot{\theta}^2 > 0$$

so $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t$ and $\mathbf{e}_b = \mathbf{k}_0$.

(d) Tangential and normal components of the acceleration

> **Pedagogical Justification:** $a_t = \dot{v}$ is the rate of change of the speed and $a_n = v^2/\rho$ measures the turning of the velocity.

$$a_t = \frac{d}{dt}\left(\sqrt{1+\beta^2}\, r\dot{\theta}\right) = \sqrt{1+\beta^2}\left(\dot{r}\dot{\theta} + r\ddot{\theta}\right) = \sqrt{1+\beta^2}\left(\beta r\dot{\theta}^2 + r\ddot{\theta}\right) = \sqrt{1+\beta^2}\, r\left[\beta n^2\delta^2 t^{2n-2} + n(n-1)\delta t^{n-2}\right] = \sqrt{1+\beta^2}\, n\delta t^{n-2}\, r\left[\beta n\delta t^{n} + (n-1)\right]$$

- **Normal component.** The cross product $\mathbf{a} \times \mathbf{v} = (a_t \mathbf{e}_t + a_n \mathbf{e}_n) \times v\, \mathbf{e}_t = -v a_n\, \mathbf{e}_b$ gives $a_n = \dfrac{\|\mathbf{v} \times \mathbf{a}\|}{v} = \dfrac{|v_r a_\theta - v_\theta a_r|}{v}$. The numerator is

$$v_r a_\theta - v_\theta a_r = \left(\beta r \dot{\theta}\right)\left(r\ddot{\theta} + 2\beta r \dot{\theta}^2\right) - \left(r\dot{\theta}\right)\left(\left(\beta^2 - 1\right) r\dot{\theta}^2 + \beta r \ddot{\theta}\right)$$

$$= \beta r^2 \dot{\theta}\ddot{\theta} + 2\beta^2 r^2 \dot{\theta}^3 - \left(\beta^2 - 1\right) r^2 \dot{\theta}^3 - \beta r^2 \dot{\theta}\ddot{\theta} = \left(\beta^2 + 1\right) r^2 \dot{\theta}^3$$

Dividing by $v = r\dot{\theta}\sqrt{\beta^2 + 1}$:

$$a_n = \frac{\left(\beta^2 + 1\right) r^2 \dot{\theta}^3}{r\dot{\theta}\sqrt{\beta^2 + 1}} = r\dot{\theta}^2 \sqrt{\beta^2 + 1} = n^2 \delta^2 t^{2n-2}\, r\sqrt{1 + \beta^2}$$

Here $\|\mathbf{v} \times \mathbf{a}\| = |v_r a_\theta - v_\theta a_r|$, because $(v_r \mathbf{e}_r + v_\theta \mathbf{e}_\theta) \times (a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta) = (v_r a_\theta - v_\theta a_r)\, \mathbf{k}_0$ ($\mathbf{e}_r \times \mathbf{e}_r = \mathbf{e}_\theta \times \mathbf{e}_\theta = \mathbf{0}, \mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}_0$).

3.2 Forces

Not applicable: kinematics-only problem. The path and the motion law are given.

3.3 Equations of Motion

Not applicable: kinematics-only problem. The motion is prescribed by $\theta(t) = \delta t^n$.

3.4 Integration and Answers

(e) Radius of curvature

> **Pedagogical Justification:** the radius of curvature follows from the normal acceleration, $a_n = v^2/\rho$:

$$\rho = \frac{v^2}{a_n} = \frac{r^2\dot{\theta}^2\left(1 + \beta^2\right)}{r\dot{\theta}^2\sqrt{1 + \beta^2}} = \boxed{r\sqrt{1 + \beta^2}}$$

Interpretation & Checks

- **Geometric invariance:** on the spiral the angle ψ between the radial line and the tangent satisfies $\tan\psi = r/(dr/d\theta) = 1/\beta$, a constant. The spiral is self-similar: $\lambda\, r(\theta) = r(\theta + \ln\lambda/\beta)$, so scaling it by a factor λ gives the same curve rotated by an angle of magnitude $\ln\lambda/\beta$. Hence every length of the curve, ρ included, scales with r.
- **Dimensional check:** $[\rho] = [r]\left[\sqrt{1+\beta^2}\right] = \mathrm{m}$; $[v_r] = [v_\theta] = [r\dot\theta] = \mathrm{m/s}$, and $[a_n] = [r\dot\theta^2] = \mathrm{m/s^2}$.
- **Limiting case $\beta \to 0$:** the spiral degenerates into the circle $r = \alpha$, with $\rho = r$ and $a_r = -r\dot\theta^2$, the centripetal acceleration.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 2 in its solution keys. The results are verified by independent derivation: sympy reproduces $v_r/(r\dot\theta) = \beta$, $v_\theta/(r\dot\theta) = 1$, the factorised forms of a_r, a_θ and a_t, the identity $a_n^2 = r^2\dot\theta^4\left(1+\beta^2\right)$ and $\rho = r\sqrt{1+\beta^2}$, with zero residual.

Source: Problems.pdf